

Predictive Modelling

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4: Model building and forecasting

Outline

Modeling cycle

Diagnostic checks

Model selection, in sample

Forecasting, hold-out sample

Forecast evaluation

Modeling cycle

Split the sample in two parts. The first part is for model creation and selection. The second part is for the evaluation of forecasts. Put both samples together, re-estimate all parameters and make forecasts for the future.

Select the elements of $g(\cdot)$. Estimate the parameters. Check if the estimated shocks ε_t are approximately white noise. Select one or two (or three) adequate models. Make recursive forecasts for second sample. Evaluate forecasts and choose one (or two) final models.

Recursive means: estimate model until time T , forecast $T + 1$. Estimate model until time $T + 1$, forecast $T + 2$, and so on.

Steps

Decide on levels or growth rates. If data are trending, or when the levels have no natural meaning (AEX, GDP), take growth rates. In case of doubt, use test to be discussed later on.

Decide on deterministic regressors (always include constant), that is, linear trend, seasonal dummies, sine and cosine functions, dummies for outliers, and dummy for structural break, if relevant. Regress the variable on these deterministics and continue with residuals as the new variable.

Decide on the possible inclusion of an additional variable z_t (with or without lags)

In short

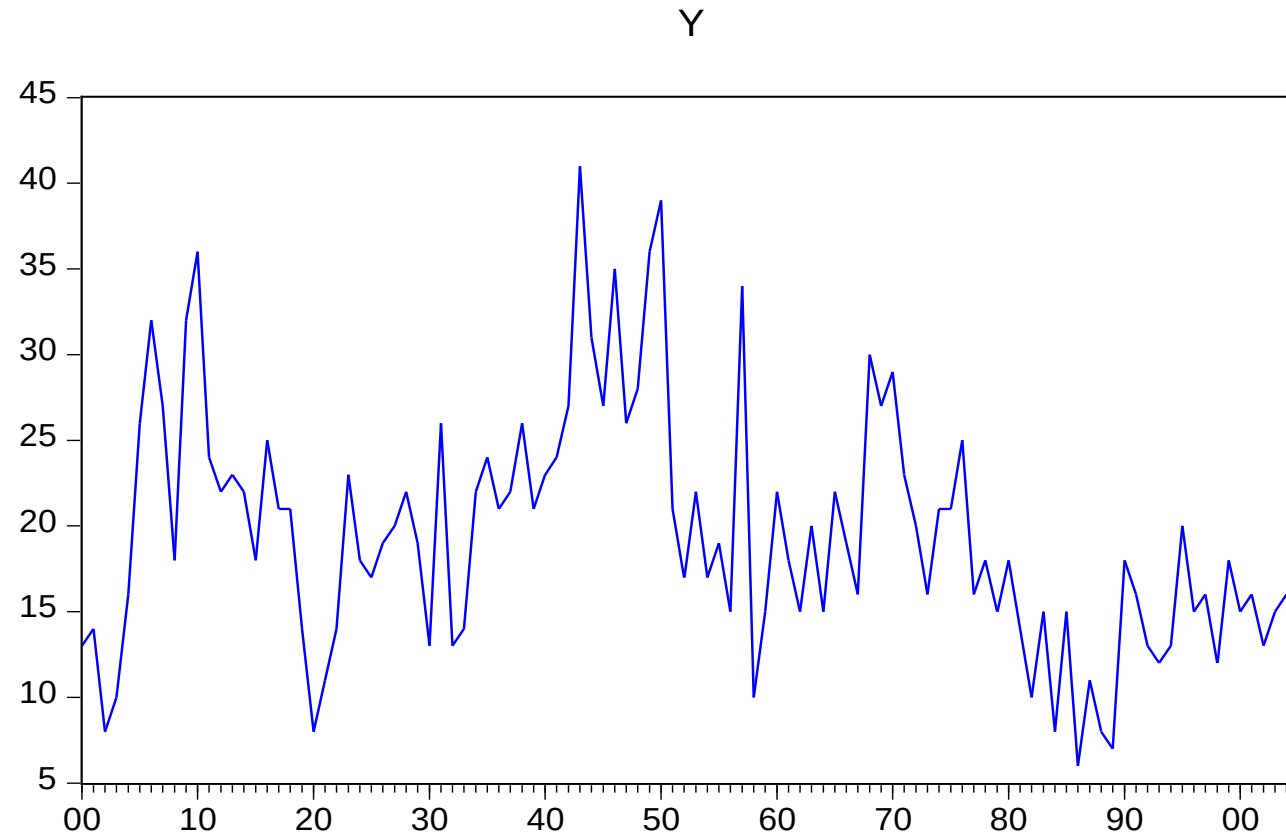
After appropriate transformation (if needed: differencing), we have y_t

$$y_t = f_1(\textit{deterministics}) + f_2(z_t) + y_t^*$$

Estimate the parameters in $f_1(\cdot)$ and $f_2(\cdot)$, and compute y_t^*

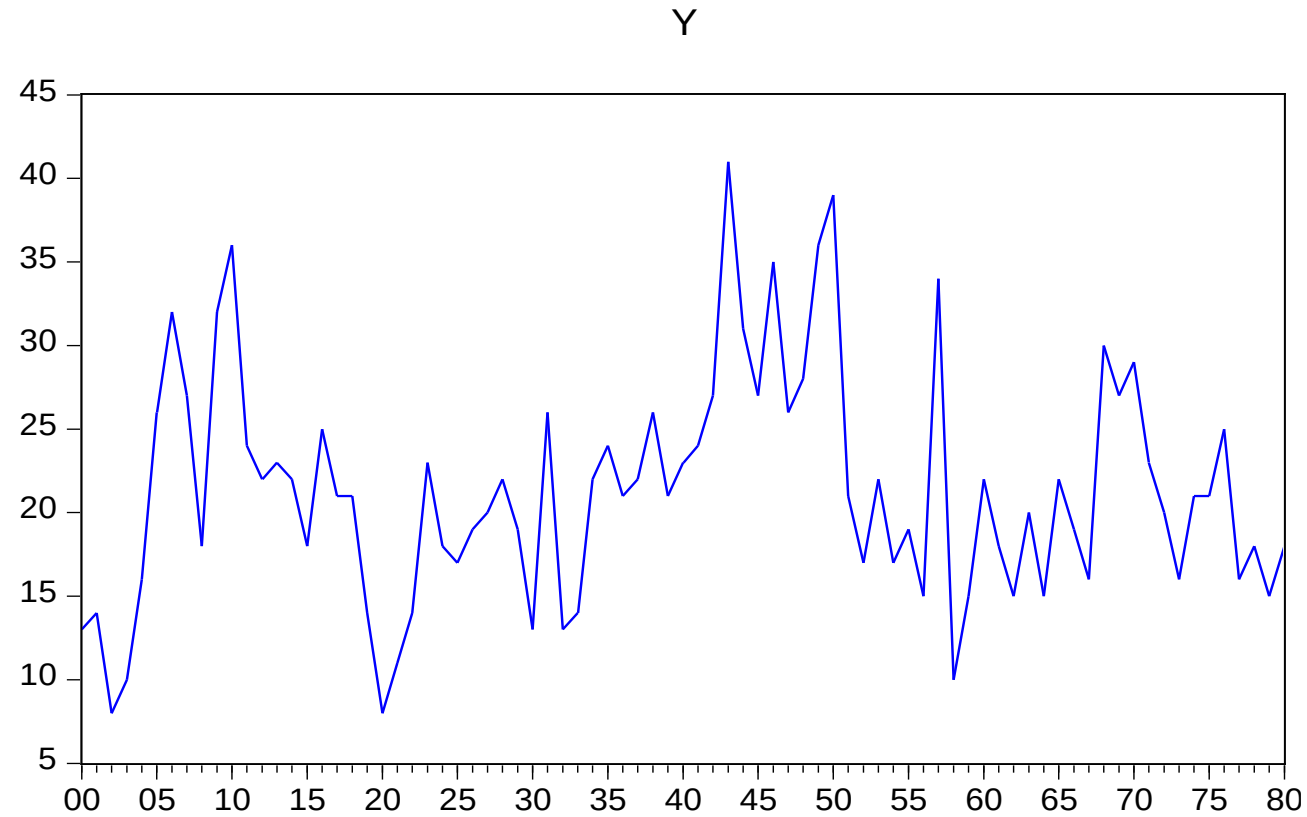
Check the (partial) autocorrelations on y_t^* and decide on p and q in ARMA models. Consider a few models.

Running example: Earthquakes, 1900-2005

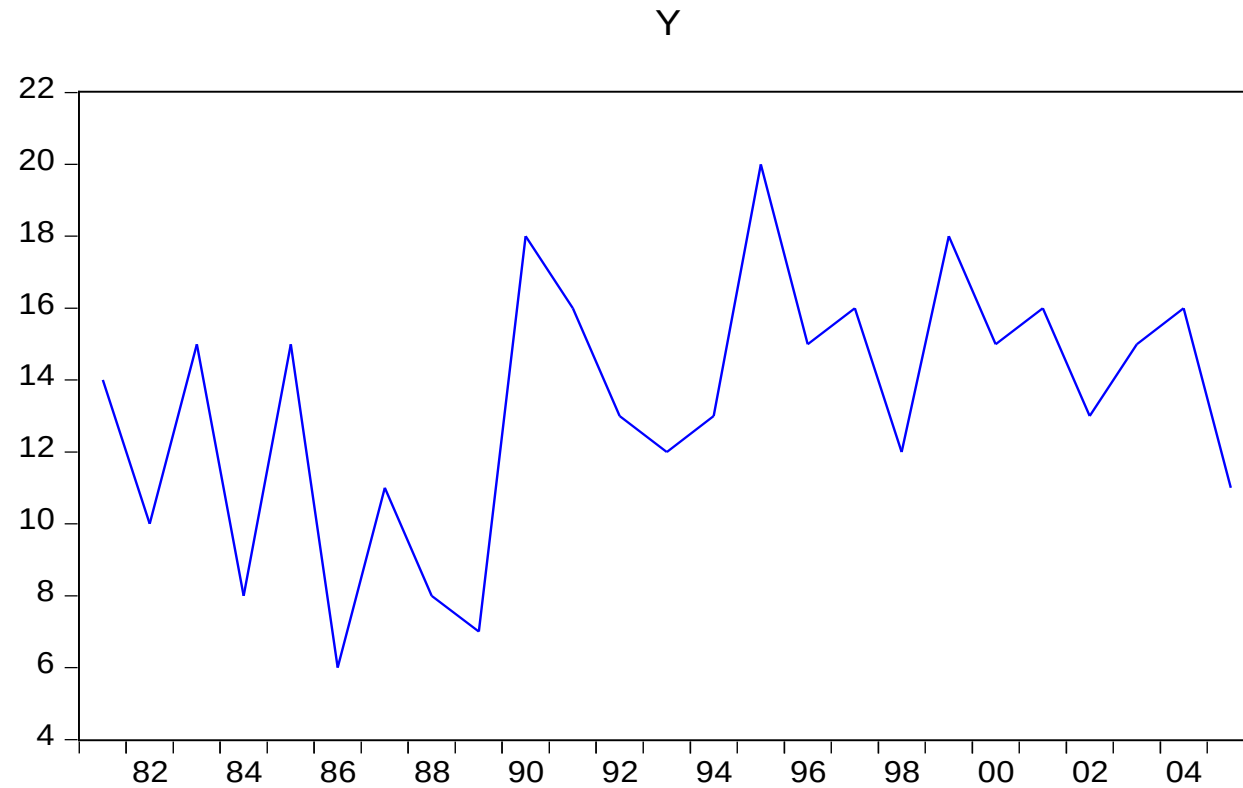


No trend, so continue with levels

Estimation sample 1900-1980



Hold-out sample (1981-2005)



(Partial) autocorrelations (T = 81)

Two times the standard error is $\frac{2}{\sqrt{81}} = 0.222$

Lag	Autocorrelation	Partial autocorrelation
1	0.471	0.471
2	0.282	0.078
3	0.269	0.143
4	0.189	0.004
5	0.080	-0.060
6	0.079	0.029
7	0.075	0.017
8	-0.039	-0.110
9	-0.199	-0.212
10	-0.230	-0.110
11	-0.106	0.114
12	-0.126	-0.009

Model selection criteria

The models should describe the variable y_t as good as possible. At the same time, we prefer small models. We should find a balance between model fit and model parsimony (number of parameters to be estimated).

The Akaike Information Criterion [AIC] is given by

$$AIC(k) = T \ln \hat{\sigma}^2 + 2k$$

where k is number of parameters and $\hat{\sigma}^2$ is the residual variance.

The Schwarz Information Criterion [SIC] is given by

$$SIC(k) = T \ln \hat{\sigma}^2 + k \ln(T)$$

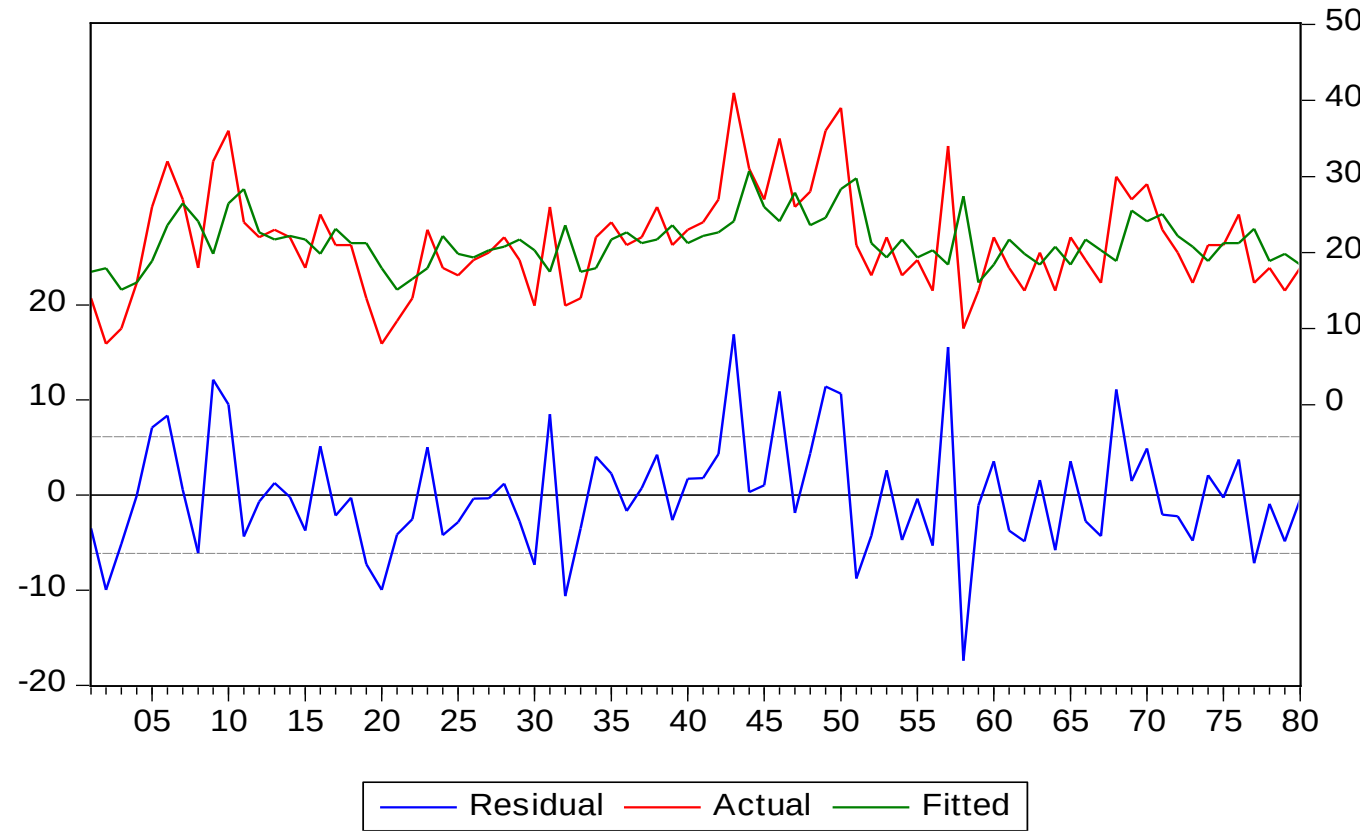
Select ARMA orders p and q that minimize $AIC(k)$ or $SIC(k)$.

AR(1) for Earthquakes

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	11.352	2.221	5.111	0.000
Y(-1)	0.472	0.099	4.773	0.000

AIC	6.490
SIC	6.550

Residuals



Diagnostic tests

One of the assumptions of white noise is

$$E(\varepsilon_s \varepsilon_t) = 0 \text{ for } s \neq t$$

Ljung Box test: Compute the autocorrelations of the estimated residuals, that is, $r_k(\hat{\varepsilon})$, and then the test is

$$LB(m) = T(T + 2) \sum_{k=1}^m \frac{r_k^2(\hat{\varepsilon})}{T - k}$$

Under the null hypothesis of no residual autocorrelation, this test is distributed as $\chi^2(m)$.

A more specific test for residual autocorrelation

Compute the estimated residuals $\hat{\varepsilon}_t$, and run the auxiliary regression

$$\hat{\varepsilon}_t = \alpha_1 y_{t-1} + \cdots + \alpha_p y_{t-p} + \alpha_{p+1} \hat{\varepsilon}_{t-1} + \cdots + \alpha_{p+r} \hat{\varepsilon}_{t-r} + v_t$$

Compute the R^2 of this regression, and compute the test statistic as TR^2 , which under the null hypothesis:

The AR(p) model is adequate

is distributed as $\chi^2(r)$, where the alternative hypothesis is

An AR($p+r$) model or an ARMA(p,r) is more adequate

Autocorrelation (until lag 2) in the residuals of the AR(1)?

Breusch-Godfrey Serial Correlation LM Test:

F-statistic	1.952	Prob. F(2,76)	0.149
Obs*R-squared	3.908	Prob. Chi-Square(2)	0.142

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-14.887	7.856	-1.895	0.062
Y(-1)	0.698	0.367	1.902	0.061
RESID(-1)	-0.764	0.392	-1.951	0.055
RESID(-2)	-0.364	0.206	-1.764	0.082

Other diagnostic checks

Assumption of white noise is $E(\varepsilon_t^2) = \sigma^2$

Consider auxiliary regression

$$\hat{\varepsilon}_t^2 = \omega + \alpha \hat{\varepsilon}_{t-1}^2 + w_t$$

Compute the R^2 of this regression, and compute the test statistic as TR^2 , which under the null hypothesis “constant variance” is distributed as $\chi^2(1)$, where the alternative hypothesis is that the variance is not constant but can be predicted from the past.

Heteroskedasticity in errors AR(1)?

In the regression

$$\hat{\varepsilon}_t^2 = \omega + \alpha \hat{\varepsilon}_{t-1}^2 + w_t$$

we have

$$\hat{\omega} = 28.960 (7.863)$$

$$\hat{\alpha} = 0.216 (0.111)$$

$$TR^2 = 3.681$$

5% $\chi^2(1)$ is 3.84

Normality (to check for extreme outliers)

Compute

$$\hat{m}_j = \frac{1}{T} \sum_{t=1}^T \hat{\varepsilon}_t^j$$

With these compute:

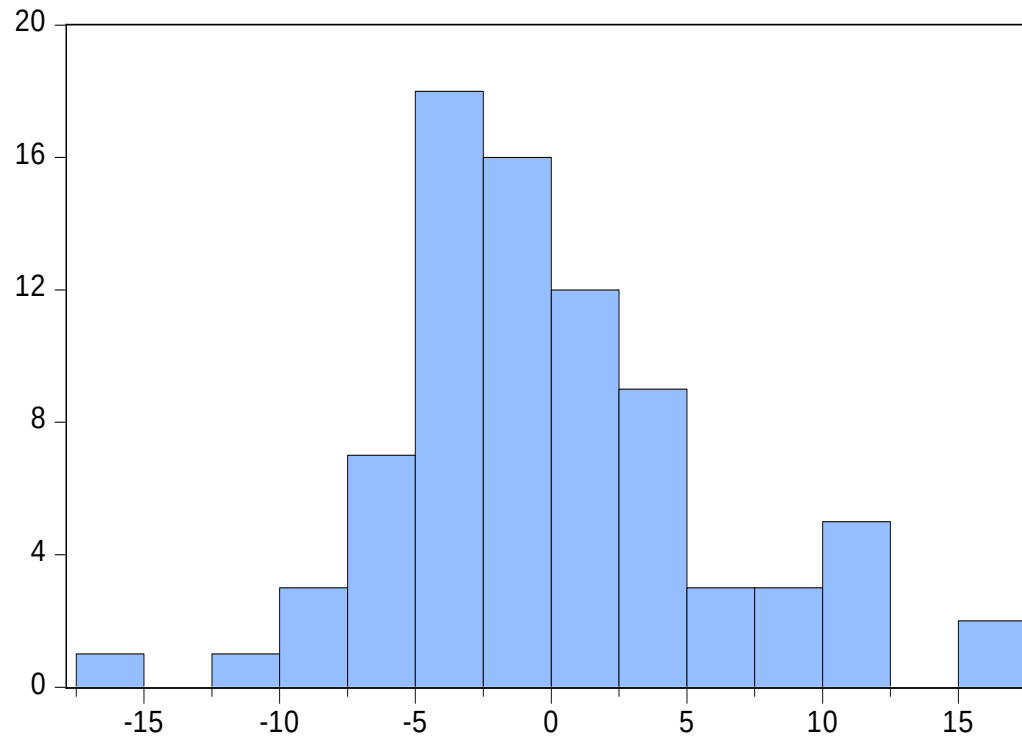
$$\text{Skewness: } SK_{\hat{\varepsilon}} = \frac{\hat{m}_3}{\sqrt{\hat{m}_2^3}}, \quad \text{Kurtosis: } K_{\hat{\varepsilon}} = \frac{\hat{m}_4}{\hat{m}_2^2}$$

Test the null hypothesis that Skewness is 0 and Kurtosis is 3, using the test statistic

$$JB = \frac{T}{6} SK_{\hat{\varepsilon}}^2 + \frac{T}{24} (K_{\hat{\varepsilon}} - 3)^2$$

Under the null hypothesis the JB test has the $\chi^2(2)$ distribution

Normality, approximately?



Series: RESID	
Sample 1900 1980	
Observations 80	
Mean	3.02e-15
Median	-0.373271
Maximum	16.90849
Minimum	-17.39427
Std. Dev.	6.094657
Skewness	0.394009
Kurtosis	3.655670
Jarque-Bera	3.502921
Probability	0.173520

Alternative models for Earthquakes: AR(2)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	10.597	2.603	4.071	0.000
Y(-1)	0.423	0.114	3.711	0.000
Y(-2)	0.086	0.114	0.756	0.452
AIC	6.517			
SIC	6.607			

Another one: ARMA(1,1)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	5.383	2.859	1.883	0.064
Y(-1)	0.752	0.132	5.703	0.000
MA(1)	-0.410	0.188	-2.178	0.032

AIC	6.487
SIC	6.576

Forecasting

For a given variable of interest, different types of forecasts may be given:

1. Point forecasts: “Inflation in the euro area over the next twelve months is expected to be equal to 2.3 percent.”
2. Interval forecasts: “Inflation in the euro area over the next twelve months will be between 1.0 and 3.6 percent with probability 0.95.”
3. Probability forecasts: “There is a probability of 0.95 that inflation in the euro area over the next twelve months is lower than inflation over the previous twelve months.”
4. Density forecasts: “Inflation in the euro area over the next twelve months is normally distributed with mean equal to 2.3 percent and standard deviation 0.65.”

Point forecasts: the AR(1) model

Consider the AR(1) model

$$y_t = \phi_1 y_{t-1} + \varepsilon_t$$

with $E(\varepsilon_t) = 0$ and $E(\varepsilon_t^2) = \sigma^2$.

$$y_{T+1} = \phi_1 y_T + \varepsilon_{T+1}$$

Because $E(\varepsilon_{T+1}) = 0$, we have

$$\hat{y}_{T+1|T} = \phi_1 y_T$$

One-step-ahead forecast error

$$e_{T+1|T} = y_{T+1} - \hat{y}_{T+1|T} = \phi_1 y_T + \varepsilon_{T+1} - \phi_1 y_T = \varepsilon_{T+1}$$

The variance of the one-step-ahead forecast error $e_{T+1|T}$ is

$$\text{Variance}(\varepsilon_{T+1}) = \sigma^2$$

Note: a positive forecast error means that the forecast was too low, that is

$$y_{T+1} > \hat{y}_{T+1|T}$$

Evaluation of P forecasts

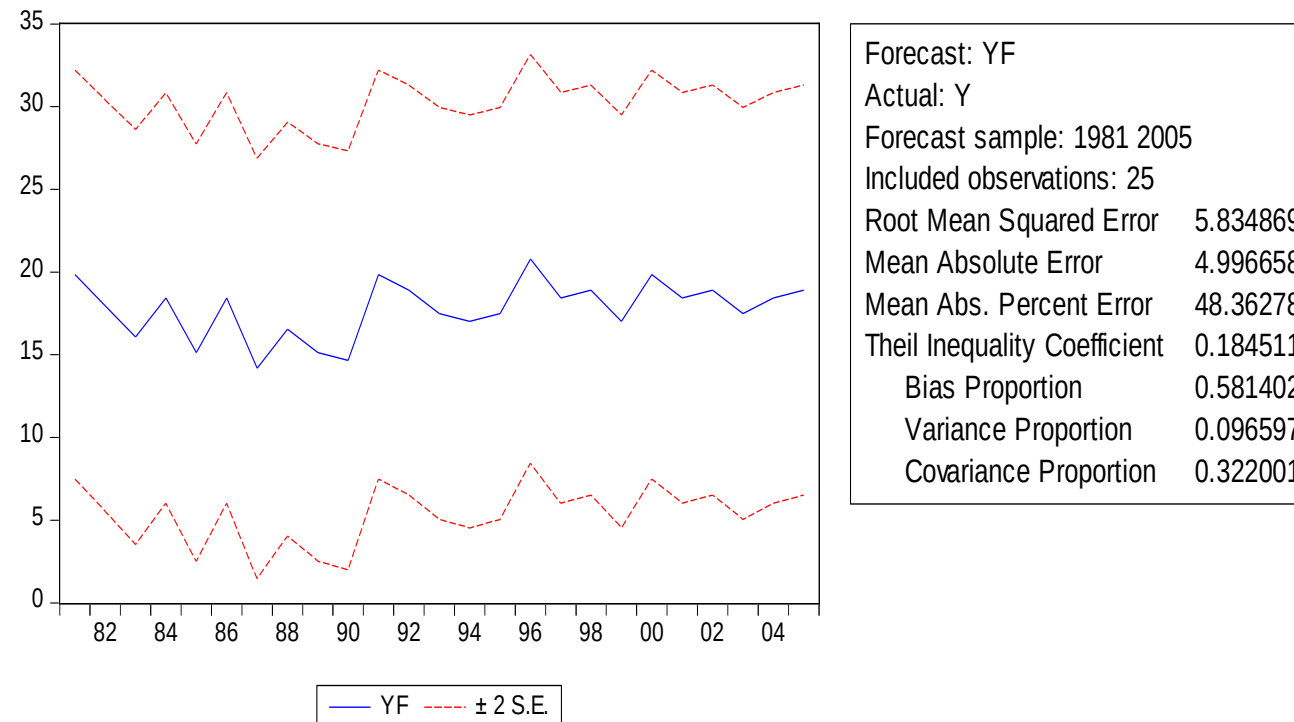
Root Mean Squared Prediction Error (RMSPE)

$$\sqrt{\frac{1}{P} \sum_{t=T}^{t=T+P-1} (y_{t+1} - \hat{y}_{t+1|t})^2}$$

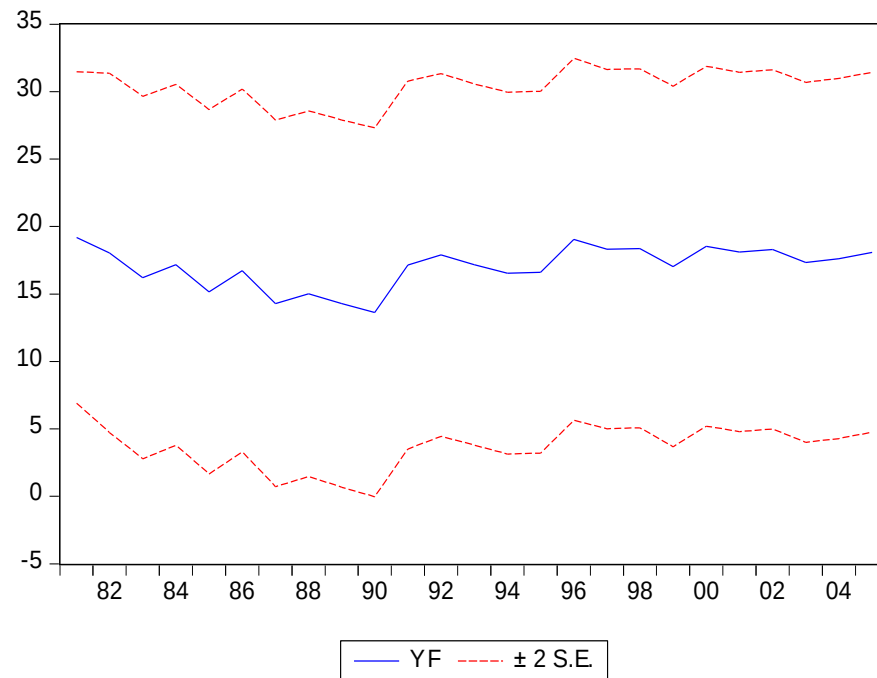
Mean Absolute Error (MAE)

$$\frac{1}{P} \sum_{t=T}^{t=T+P-1} |y_{t+1} - \hat{y}_{t+1|t}|$$

Recursive 1-step-ahead forecasts for AR(1) plus/minus 2 times σ

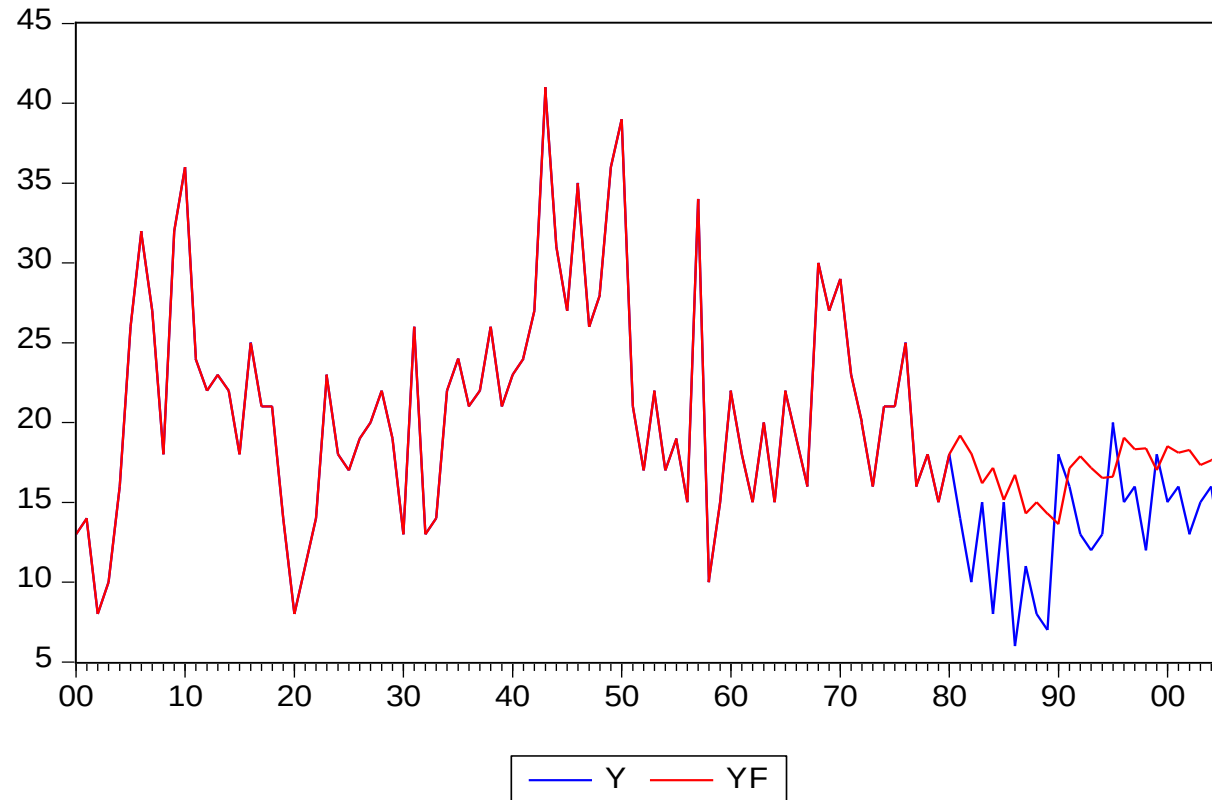


Recursive 1-step-ahead forecasts for ARMA(1,1)



Forecast: YF
Actual: Y
Forecast sample: 1981 2005
Included observations: 25
Root Mean Squared Error 5.163801
Mean Absolute Error 4.409641
Mean Abs. Percent Error 42.60750
Theil Inequality Coefficient 0.167274
Bias Proportion 0.516410
Variance Proportion 0.151500
Covariance Proportion 0.332091

Forecasts from ARMA(1,1), holdout-sample



Two-steps-ahead forecasts for an AR(1)

At time $T + 2$ we have

$$y_{T+2} = \phi_1 y_{T+1} + \varepsilon_{T+2} = \phi_1^2 y_T + \phi_1 \varepsilon_{T+1} + \varepsilon_{T+2}$$

Hence, the forecast from time T is

$$\hat{y}_{T+2|T} = \phi_1^2 y_T$$

The forecast error is

$$e_{T+2|T} = \phi_1 \varepsilon_{T+1} + \varepsilon_{T+2}$$

The variance of the two-steps-ahead forecast error is

$$V(e_{T+2|T}) = (1 + \phi_1^2) \sigma^2$$

Thus,

$$V(e_{T+2|T}) > V(e_{T+1|T})$$

Forecasts from an ARMA(1,1)

$$y_t = \phi_1 y_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

At time $T + 1$, the observation is

$$y_{T+1} = \phi_1 y_T + \varepsilon_{T+1} + \theta_1 \varepsilon_T$$

So the forecast is

$$\hat{y}_{T+1|T} = \phi_1 y_T + \theta_1 \varepsilon_T$$

And the forecast error is ε_{T+1} .

Two steps ahead, we have

$$\begin{aligned} y_{T+2} &= \phi_1 y_{T+1} + \varepsilon_{T+2} + \theta_1 \varepsilon_{T+1} \\ &= \phi_1^2 y_T + \phi_1 \varepsilon_{T+1} + \phi_1 \theta_1 \varepsilon_T + \varepsilon_{T+2} + \theta_1 \varepsilon_{T+1} \end{aligned}$$

The forecast is

$$\hat{y}_{T+2|T} = \phi_1^2 y_T + \phi_1 \theta_1 \varepsilon_T = \phi_1 \hat{y}_{T+1|T}$$

The forecast error is $\varepsilon_{T+2} + (\phi_1 + \theta_1) \varepsilon_{T+1}$

Three-steps-ahead forecasts for an AR(1)

True observation is

$$y_{T+3} = \phi_1^3 y_T + \phi_1^2 \varepsilon_{T+1} + \phi_1 \varepsilon_{T+2} + \varepsilon_{T+3}$$

Forecast is

$$\hat{y}_{T+3|T} = \phi_1^3 y_T$$

Variance for three steps ahead is

$$V(e_{T+3|T}) = (1 + \phi_1^2 + \phi_1^4) \sigma^2$$

Multiple-steps-ahead forecasts from AR(1)

$$V(e_{T+h|T}) = (1 + \phi_1^2 + \phi_1^4 + \dots + \phi_1^{2(h-1)})\sigma^2$$

When $h \rightarrow \infty$, and $|\phi_1| < 1$,

$$V(e_{T+h|T}) \rightarrow \frac{\sigma^2}{1 - \phi_1^2} = \gamma_0$$

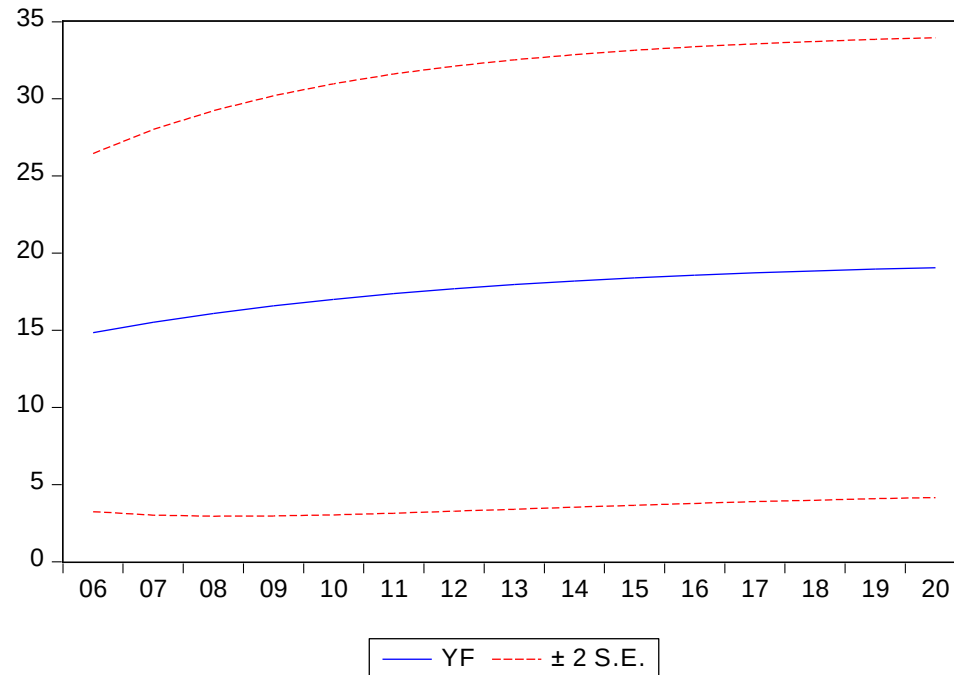
With increasing h , the forecast error variance approaches the unconditional variance. This gives an insight in how well and how far ahead a time series can be predicted.

Forecasts for Earthquakes from ARMA(1,1) from 2005 onwards

Re-estimating the parameters for the full sample

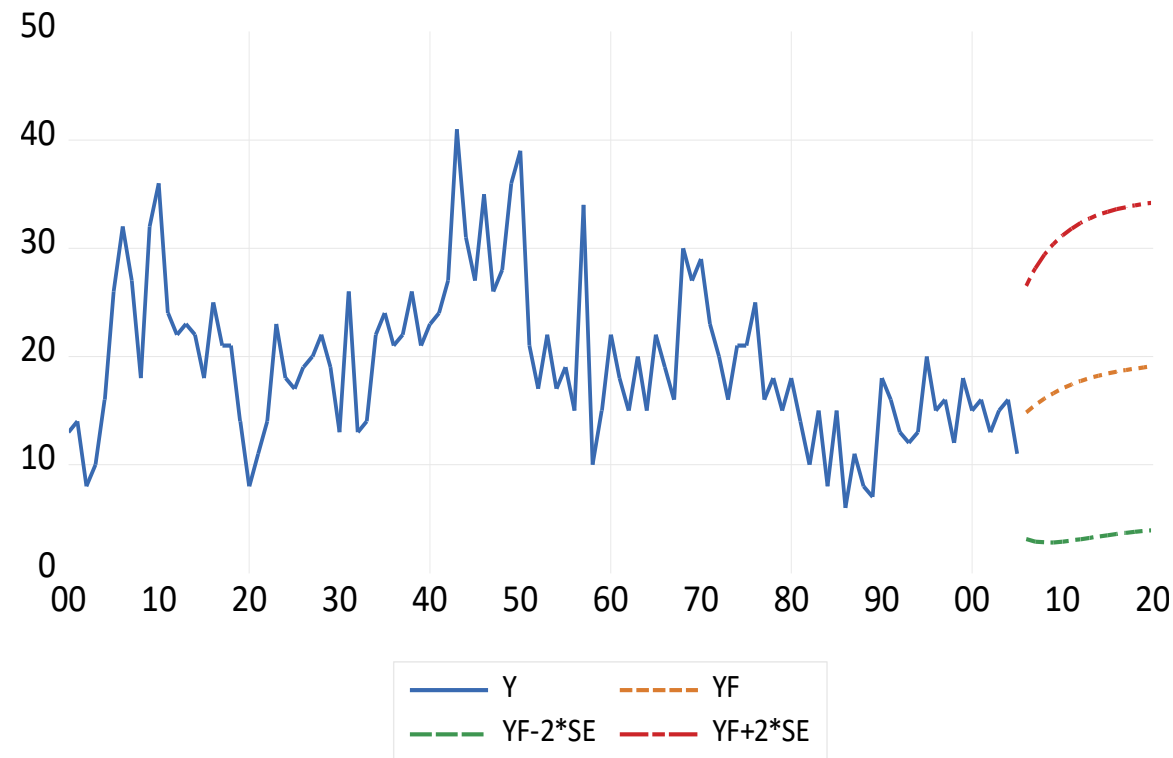
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.785	1.438	1.937	0.056
Y(-1)	0.858	0.070	12.191	0.000
MA(1)	-0.471	0.120	-3.936	0.000

Forecasts from ARMA(1,1) from 2005 onwards



Perhaps only one-step-ahead or two-steps-ahead forecasts are useful

Forecasts from ARMA(1,1) from 2005 onwards



The forecast distribution quickly converges to the unconditional distribution

Can we use direct multi-step estimation and forecasting?

Perhaps it is useful to consider

$$y_t = \mu + \alpha y_{t-k} + \varepsilon_t$$

for $k=1,2,3,4, \dots$, and directly apply OLS and create forecasts

Research shows that this may work when (i) the model is misspecified and (ii) the sample is small.

Earthquake data again

Sample 1908 to 1990, dynamic forecasts (all horizons at the same time) and static forecasts, each time with the most recent information included. Root Mean Squared Prediction Error for the sample 1991 to 2005

	Estimated α (with standard error)	Dynamic RMSPE	Static RMSPE
$k = 1$	0.521 (0.094)	5.889	4.044
$k = 2$	0.430 (0.101)	5.248	4.175
$k = 3$	0.468 (0.101)	4.745	4.015
$k = 4$	0.408 (0.105)	4.506	3.551
$k = 5$	0.294 (0.111)	4.710	4.066
$k = 6$	0.219 (0.111)	5.182	4.752
$k = 7$	0.202 (0.113)	5.115	4.743

What did we learn?

Modeling cycle

Diagnostic checks (to see if estimated residuals are approximately white noise)

Model selection, in sample (retain two or three models for the hold-out sample)

(Recursive) forecasts in hold-out sample, choose winner

Forecast out-of-sample with intervals, decide on reasonable horizon where forecasts are helpful.